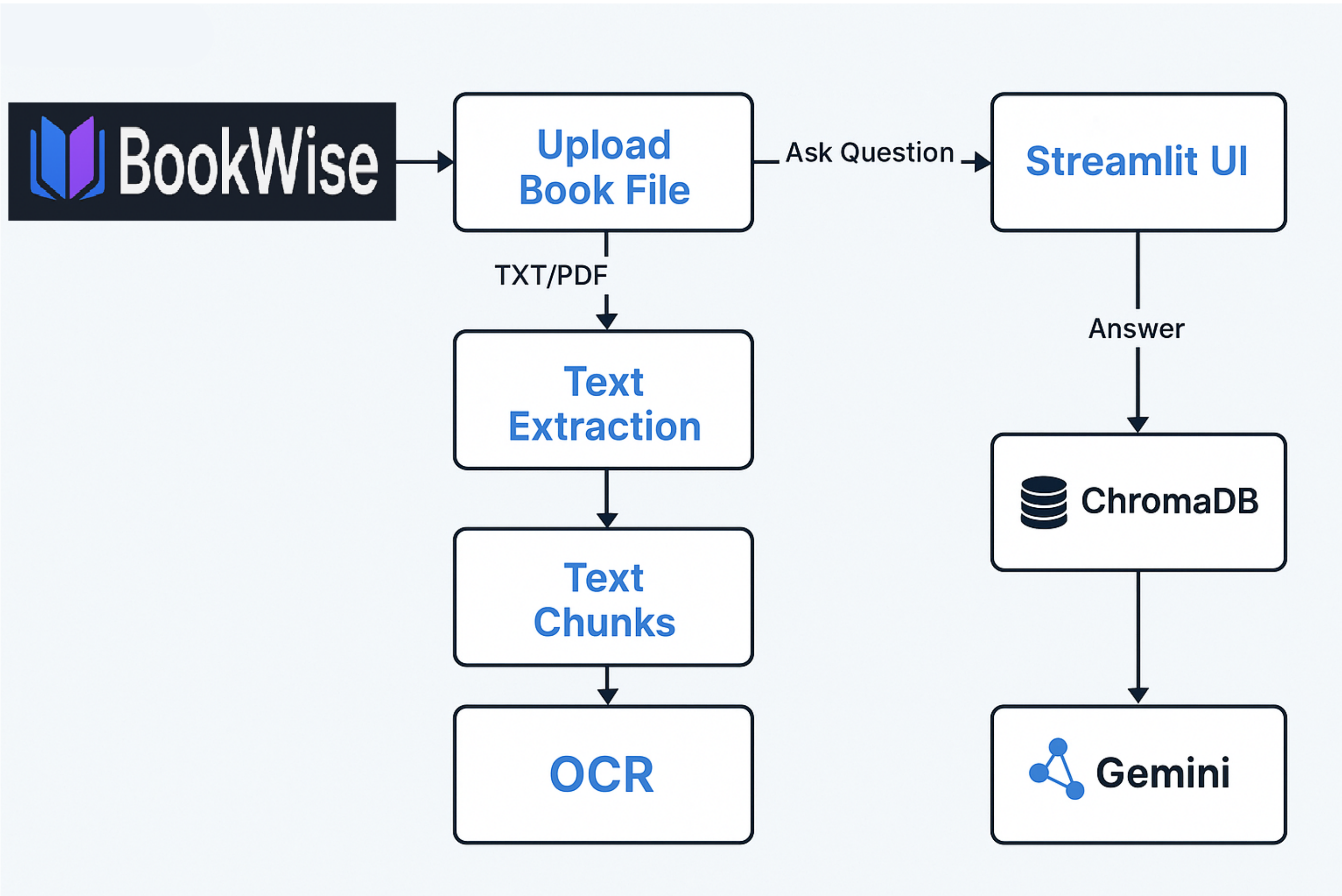
BookWise - Ask Your Books AI



Introduction

BookWise is an AI-powered platform that lets users upload books in TXT, PDF, or image formats and ask natural language questions. It uses Google Gemini to provide intelligent answers and ChromaDB for fast semantic search.

Objective

Enable users to interact with their books as if they were intelligent, searchable databases - helping students, readers, and researchers get concise insights from chapters, scanned pages, or entire books.

Key Features

- Upload .txt, .pdf, or image files (JPG/PNG)
- Extract text using OCR and PDF parsing
- Chunk and embed content into ChromaDB
- Retrieve semantically similar chunks using sentence-transformers
- Generate responses with Google Gemini
- Maintain session memory to track Q&A
- Detect chapters for better structure-aware answering

Architecture Overview

The architecture includes:

1. Streamlit frontend with dark mode for user interaction.
2. Flask backend handles embedding and LLM-based answering.
3. OCR (Tesseract) and PDFPlumber for extracting text from uploaded content.
4. ChromaDB as a vector store for semantic retrieval.
5. Google Gemini LLM for response generation based on retrieved chunks.

Below is the architectural diagram for BookWise:

Refer to the visual diagram below for a high-level view of the BookWise system.

Chapter Detection

BookWise detects chapters using regular expressions like 'Chapter 1', 'CHAPTER II', etc., and segments books accordingly. This enables chapter-specific queries such as 'Summarize Chapter 3'.

Session Memory

The app stores user questions and AI responses during a session using Streamlit's session state. It enables users to revisit their previous queries in a single sitting, just like a chat.

Setup Instructions

1. Unzip the project
2. Create a virtual environment and install dependencies using `pip install -r requirements.txt`
3. Add your Google API key to `.env`
4. Run the app using `streamlit run ui/streamlit_app.py`
5. Upload a book and ask questions!

Challenges

- Ensuring clean OCR from noisy images
- Detecting chapters reliably across various formatting styles
- Balancing chunk size vs context for better answers

Future Enhancements

- Bookmark responses
- Multi-turn chat history persistence
- Audio read-out of summaries
- Custom document tagging and classification

Conclusion

BookWise shows how generative AI + semantic search can revolutionize how we interact with long-form content like books and study material.